# A New Context-Aware Framework for Defending Against Adversarial Attacks in Hyperspectral Image Classification

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***Abstract*— Deep neural networks play a significant role in hyperspectral image (HSI) processing, yet they can be easily fooled when trained with adversarial samples (generated by adding tiny perturbations to clean samples). These perturbations are invisible to the human eye, but can easily lead to misclassification by the deep learning model. Recent research on defense against adversarial samples in HSI classification has improved the robustness of deep networks by exploiting global contextual information. However, available methods do not distinguish between different classes of contextual information, which makes the global context unreliable and increases the success rate of attacks. To solve this problem, we propose a robust context-aware network able to defend against adversarial samples in HSI classification. The proposed model generates a global contextual representation by aggregating the features learned via dilated convolution, and then explicitly models intraclass and interclass contextual information by constructing a class context-aware learning module (including affinity loss) to further refine the global context. The module helps pixels obtain more reliable long-range dependencies and improves the overall robustness of the model against adversarial attacks. Experiments on several benchmark HSI datasets demonstrate that the proposed method is more robust and exhibits better generalization than other advanced techniques.**

***Index Terms*— Adversarial attacks, adversarial samples, deep learning, global context, hyperspectral images (HSIs).**

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## I. Introduction

DUE to the rapid development of satellite sensor technology, remote sensing images acquired by Earth observation (EO) instruments have evolved from panchromatic to multispectral and then to hyperspectral with hundreds of continuous wavebands. Hyperspectral images (HSIs) are rich in spectral information and can be used to characterize the physical properties of ground objects more accurately than other imaging approaches. As a basic task, HSI classification aims to assign a unique label to each pixel. It is widely used in many remote sensing applications, including military reconnaissance, agricultural production, environmental monitoring, and ore exploration, among many others [1], [2], [3], [4].

The rapid development of deep or machine learning has lead to new remote sensing processing methods [5], [6], [7], [8]. Chen et al. [9] used principal component analysis (PCA) to reduce the dimensionality of HSIs, and then applied a stacked autoencoder (SAE) to extract features from the HSI. Hang et al. [10] designed a cascaded recurrent neural network (RNN) to explore redundant and complementary information in HSIs. Long and short-term memory (LSTM) neural networks have also been widely used in HSI classification tasks [11]; however, their main focus is on the spectral information contained in the HSI. To extend this approach to the spectral–spatial domain, Liu et al. [12] designed a bidirectional convolutional LSTM (ConvLSTM) to learn both the spectral and spatial features. Wang et al. [13] combined the ConvLSTM and a generative adversarial capsule network to address the problem of insufficient sample size in deep learning.

Convolutional neural networks (CNNs) have also been widely used in HSI classification tasks due to their powerful fitting ability [14], [15], [16], [17]. Zheng et al. [18] designed a rotation-invariant attention network to improve the classification performance of HSI under spatial rotation. To make full use of the limited samples in HSI, Yao et al. [19] used superpixel segmentation to propose a semi-supervised CNN framework based on active learning. The performance of these methods demonstrated the effectiveness of CNNs in feature extraction tasks. To further improve HSI interpretation, other methods combining CNNs have been explored. The CNN-based spectral–spatial dual-branch networks have also

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become a research focus. These networks include a combination of CNNs and RNNs or its variants [20], [21], [22]. Hong et al. [23] explored the fusion of graph convolutional neural networks (GCNs) with CNNs for HSI classification and proposed a mini-batch GCN to reduce the high computational cost incurred by constructing adjacency matrices. These methods usually have separate spectral and spatial feature streams, and the obtained spectral–spatial feature vectors are fused to produce the final results. In addition, the transformer (which has gained considerable attention in the field of natural language processing) has also been extensively explored in HSI processing in combination with CNNs [24], [25], [26].

Although the above methods achieved excellent performance, they still approach the classification problem using pixel patches rather than the whole image, an approach which is not conducive to characterizing the nonlocal dependence of pixels in spatial terms. Recent studies treat HSI classification as a pixel-level segmentation problem [27], [28]. Xu et al. [29] proposed a fully convolutional network (FCN) classification framework which takes the whole HSI as the input and produces a classification result map with the same spatial size as the ground truth. Zheng et al. [30] designed an end-to-end fast patch-free framework for HSI classification, which can improve the global spatial information mining capability of model. In [31], an end-to-end full convolutional segmentation network was proposed for solving HSI classification problems with complex spatial distributions.

The above deep-learning-based methods have shown excellent performance in HSI processing, but they normally deal with images that have not been corrupted. In fact, deep learning models are very vulnerable to adversarial attacks. A graphical illustration of an adversarial attack against a deep learning model is given in Fig. 1. As shown in the figure, models that achieve excellent performance on clean samples may perform very poorly on adversarial samples. Adding perturbations to the original input can produce adversarial samples, and deep networks tend to have high confidence in the wrong category when classifying such samples. From a human visual perspective, there are no differences between adversarial samples and clean samples. The existence of this phenomenon limits the practical applicability of deep learning models in some fields with high security requirements, such as in military and medical diagnosis applications [32].

In the field of remote sensing, Czaja et al. [33] first investigated the effects of adversarial attacks on remote sensing image classification, and the results indicated that classification models generally exhibit low robustness to adversarial samples. In [34], the damage caused by adversarial samples to the classification of remote sensing image scenes was systematically analyzed, and a training strategy to defend against adversarial attacks was developed to improve the robustness of the network. Some adversarial defense strategies in synthetic aperture radar (SAR) have also been extensively studied [35], [36], [37]. In addition, the threat of adversarial attacks in HSI processing also deserves attention. As can be seen from Fig. 1, the spectral and spatial variations produced by an adversarial attack in an HSI are imperceptible to the human eye, but these changes can greatly affect the classification performance

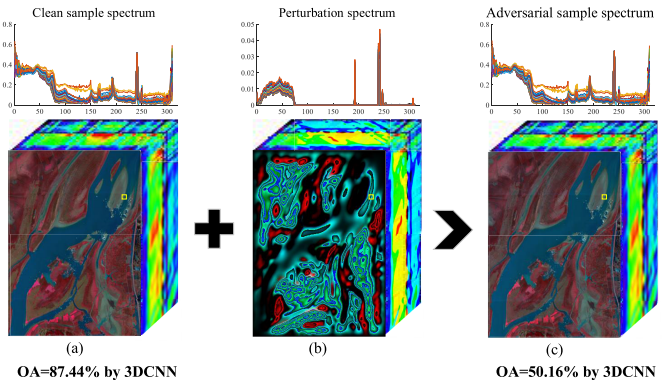


Fig. 1. Examples of adversarial sample generation in HSI classification. (a) Clean GaoFen-5 (GF-5) HSI dataset, (b) adversarial perturbations generated by the FGSM with $\epsilon = 0.1$, and (c) adversarial GF-5 HSI dataset. As it can be seen, the spectral and spatial differences between the adversarial samples and the clean samples cannot be easily perceived by the human eye. However, the deep learning model is vulnerable to adversarial attacks: taking the 3-DCNN [39] as an example, the OA decreased by 37.28% when using adversarial samples.

of a deep-learning-based model. The impact of adversarial samples in HSI was first made explicit in [38], and these results show that most advanced HSI classification methods are easily fooled by adversarial samples. Park et al. [40] proposed a framework which combines random spectral sampling and spectral shape feature encoding to improve the robustness of HSI classification, making full use of the rich spectral information present in HSIs. An improved DeepFool algorithm was proposed in [41], which deals with adversarial attacks in HSIs by adding generated boundary adversarial samples to the original training set.

The aforementioned studies illustrate that adversarial training can effectively improve the robustness of a network, but it also inevitably increases the training burden. In adversarial training, the trained model is attacked repeatedly by unknown adversarial samples [32]. Designing a robust network structure to improve the ability of the model for coping with adversarial samples is a more suitable defense strategy. Xu et al. [38] improve the defensibility of the model under adversarial attacks by capturing the global contextual information of the pixels. The established global contextual relationships can help assigned pixels share losses with other pixels, thus improving the robustness of the model. However, the global context obtained in this way is mixed due to the lack of explicit regularization. It focuses on the contextual dependencies of the same class and ignores the differences in different classes (see Fig. 2). As a result, it may include unreliable contextual information in boundary or mixed scenarios and undoubtedly increase the success rate of attack.

To overcome these limitations, we introduce a robust class context-aware (RCCA) network specifically designed to deal with adversarial samples in HSI classification. The network distinguishes between intraclass and interclass contextual information by constructing a learnable affinity matrix containing intraclass and interclass relationships that are used as class-contextual prior information. The whole network consists of three main phases. First, a backbone (based on dilated convolution) is used to generate feature maps. Second, an aggregation module is designed to aggregate the spatial–spectral

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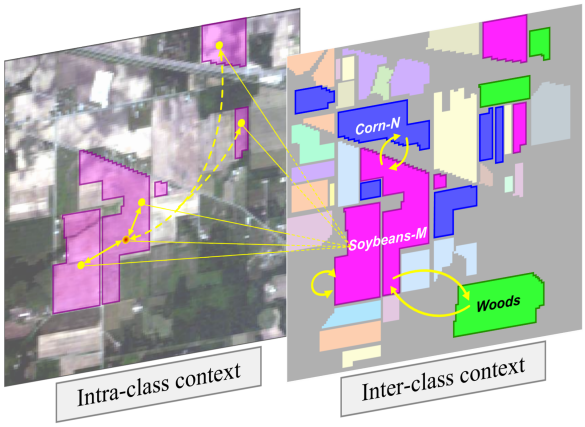


Fig. 2. Graphical illustration of intraclass contextual (the correlation of the same class) and interclass contextual (the difference of different classes) relationships in HSIs, which make the features more robust and conducive to image interpretation.

information. In the third phase, we generate the corresponding affinity prediction matrix based on the aggregated features and produce a class contextual prior (with affinity loss) to supervise the learned intraclass and interclass relationships in the affinity matrix. Using the learned class contextual prior information, we can effectively enhance the intraclass similarity and interclass variability, so that the captured global contextual information is more reliable and beneficial for the model to defend against adversarial samples. The main contributions of the article can be summarized as follows.

1) Unlike the existing context module, we introduce an affinity learning (AL) module including class priors to identify intraclass context (homogeneous correlation) and interclass context (heterogeneous variability), which facilitates enhancing the global understanding of the scene during feature learning.
2) We propose RCCA to defend against adversarial attacks in HSI classification. Compared with other HSI defense models, the proposed method improves the inherent robustness of the network by establishing more reliable global contextual relationships, which facilitates resistance to higher intensity adversarial attacks.
3) Qualitative and quantitative experiments show that the proposed model has advanced performance in different attack methods.

The remainder of this article is structured as follows. Section II reviews some related work. Section III describes the proposed RCCA method in detail. Section IV describes our experimental results and analyses, and a practical application is presented in Section V. Section VI concludes the article with some remarks and provides hints at plausible future research lines.

## II. Related Work

### A. Adversarial Attacks Methods

In this section, we briefly review the technical details of several representative white-box attack methods which can be used against deep learning algorithms.

*1) Fast Gradient Sign Method (FGSM):* To achieve effective adversarial training and improve the robustness of neural networks against adversarial samples, an attack method named FGSM was proposed in [42]. The core idea of this model is to use gradient ascent in the loss function, so that the model makes wrong decisions. For an image $I$ and a target label $\hat{y}$, an adversarial sample $I'$ generated by FGSM can be represented as follows:

$$I' = \text{clip}(I - \epsilon \cdot \text{sign}(\nabla_I J(\theta, I, \hat{y}))) \tag{1}$$

where $\epsilon$ is a value used to restrict the norm of the perturbation, $\nabla_I J(\theta, I, \hat{y})$ denotes the gradients of the objective function with input sample $I$, clip $(\cdot)$ denotes the clipping function, and sign $(\cdot)$ is the sign function. By modifying the value of $\epsilon$, we can generate adversarial samples with different perturbation strengths.

*2) Projected Gradient Descent (PGD):* As an enhanced version of FGSM, the PGD [43] improves the effectiveness of the attack by iterating the optimizer several times with smaller step sizes and projects the learned adversarial samples from each iteration into the prescribed range. Specifically, given an image $I$ and its true label $\hat{y}$, the generated adversarial sample for each iteration can be expressed as follows:

$$I'_{t+1} = \text{clip}(I'_t - \epsilon \cdot \text{sign}(\nabla_I J(\theta, I_t, \hat{y}))) \tag{2}$$

where $I'_t$ denotes the adversarial sample generated at iteration $t$, and when $t = 0$, $I'_t$ is the clean sample $I$.

*3) Carlini and Wagner (C&W):* The C&W is an optimization-based attack method [44] whose objective is to make the perturbation as small as possible (while ensuring that the model is cheated). The process of optimizing the perturbation by this method can be expressed as follows:

$$\min \|I - I'\|_2^2 + c \cdot f(I', \hat{y}) \quad \text{s.t. } I' \in [0, 1]^m \tag{3}$$

where $f(I', \hat{y})$ represents the loss function when the input is $(I, y)$, and $c$ is a hyperparameter related to attack performance.

### B. Defense Methods

A large number of defense methods have been developed to overcome the threat of adversarial attack on deep learning models, which can be divided into three main methods. The first effective defense method is adversarial training, which uses generated adversarial samples along with clean samples to train the network, thus improving the robustness of the network [45]. Cheng et al. [46] trained classifiers with adversarial samples generated during image reconstruction along with clean samples to address the adversarial attacks in remote sensing image scene classification. Although adversarial training can improve the robustness of the model to adversarial samples, it also incurs additional training burden by increasing the number of training samples, and it is a nonadaptive strategy [32].

Another defense strategy is to directly detect potential adversarial samples. This strategy focuses on perturbation detection and removal of input samples in the testing phase to reduce the impact of adversarial samples on the model [47]. Deng et al. [48] proposed a lightweight Bayesian neural network to detect possible perturbations in the input. A profiling-based perturbation detection method is proposed in [49], which

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observes both adversarial and clean samples by aggregating the effective paths of each image into class-level effective paths.

Modifying the structure of the target network is also an effective strategy to address adversarial samples and can improve the intrinsic robustness of the model [38]. Samangouei et al. [50] exploited the representation capability of GANs to improve the defense of these networks against adversarial attacks. Similarly, Jin et al. [51] proposed a defense framework based on a GAN to eliminate the adversarial perturbations of the input images. In [52], a uniform gradient regularization method was developed to penalize the generated gradients of the adversarial samples in the loss function. In this article, we adopt a strategy based on modifying the network structure to address adversarial attacks and propose an RCCA network for HSI classification which distinguishes intraclass and interclass context to improve the robustness of the model on adversarial samples.

## III. Proposed Method

The limited size with receptive field of convolution kernel limits the ability of pixels to capture long-range information, a limitation which reduces the robustness of the model against adversarial samples [53]. To address this issue, the model should capture global contextual information by establishing relationships between pixels and long-range locations [38]. However, the contextual relationships established in this way are not clearly normalized and inevitably introduce contextual relationships of a different class from the given pixel, a situation which will affect the prediction accuracy. To obtain more reliable global contextual relationships and improve the resistance of the model to adversarial attacks, we propose an RCCA network, which models relationships between classes as prior knowledge and further refines the global context by distinguishing between intraclass and interclass contextual relationships, thus improving the reliability of long-range relationships between pixels.

The proposed model consists of three main phases. In the first phase, we use dilated convolution for contextual feature extraction, which consists of three dilated convolution layers with different dilation rates and two pooling layers. In the second phase, we model the context of the obtained features using a feature aggregation (FA) module, which consists of parallel global and local context calibrators. In the third phase, we introduce an AL module to learn the intraclass and interclass contextual relationships of the captured contextual features, which will obtain reliable contextual relationships for the final classification. In the following, we provide a detailed description of the proposed RCCA method, including FA and AL modules.

### A. Feature Aggregation

To obtain effective contextual features, we designed an FA module to aggregate the features learned in phase 1 in terms of both global and local context. The FA module includes two parallel context calibrators, which are illustrated in Fig. 3.

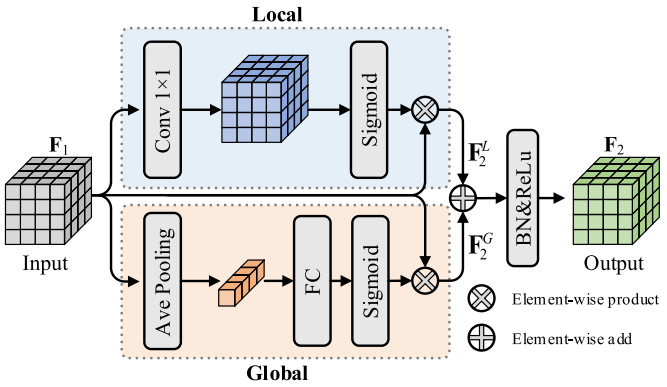


Fig. 3. Structure of the propose FA module, which includes two parallel global and local contextual feature calibrators.

*1) Local Aggregation:* The local aggregation branch focuses on the local neighborhood of pixels. For the input feature $\mathbf{F}_1 \in \mathbb{R}^{h_0 \times w_0 \times c_0}$, we first apply a convolution of size $c_0 \times 1 \times 1$ for local operation, then apply a sigmoid function to the features obtained to estimate the weights, and finally perform elementwise product operations with the original input feature map $\mathbf{F}_1$. The process can be expressed as follows:

$$\mathbf{F}_2^L = \sigma(\mathrm{Conv}_{1\times1}(\mathbf{F}_1)) \otimes \mathbf{F}_1 \tag{4}$$

where $\mathbf{F}_2^L \in \mathbb{R}^{h_0 \times w_0 \times c_0}$ denotes the output of the local aggregate branch, $\sigma$ is the sigmoid function, and $\otimes$ denotes an elementwise product.

*2) Global Aggregation:* The global aggregation branch updates the feature importance from the global context. To achieve this update, we first perform global average pooling on the input features $\mathbf{F}_1$, and then feed the processed 1-D vector into the fully connected layer (to encourage feature interactions). Finally, we use the sigmoid function to estimate the channel weights. After obtaining the weight vector, we use element replication to make the size of the resulting vector consistent with $\mathbf{F}_1$ and apply the expanded vector to the original input as follows:

$$\mathbf{F}_2^G = \mathrm{Expand}(\sigma(\mathrm{FC}(\mathrm{GAP}(\mathbf{F}_1)))) \otimes \mathbf{F}_1 \tag{5}$$

where $\mathbf{F}_2^G \in \mathbb{R}^{h_0 \times w_0 \times c_0}$ denotes the output of the global aggregate branch. FC and GAP denote the fully connected layer and the global average pooling operations, respectively. After obtaining the outputs of the two parallel branches, the final output $\mathbf{F}_2 \in \mathbb{R}^{h_0 \times w_0 \times c_0}$ is obtained by fusing these outputs as follows:

$$\mathbf{F}_2 = \delta\left(\mathrm{BN}\left(\mathbf{F}_2^L \oplus \mathbf{F}_2^G\right)\right) \tag{6}$$

where $\delta$ is the ReLU function, BN denotes the batch normalization layer, and $\oplus$ denotes an elementwise addition.

### B. Affinity Learning

To better capture the intra- and interclass contexts, inspired by [54] we introduce class context AL to improve the reliability of global context estimation. Specifically, we generate affinity predictions and corresponding affinity labels for the feature map $\mathbf{F}_2$ from phase 2. The affinity prediction includes

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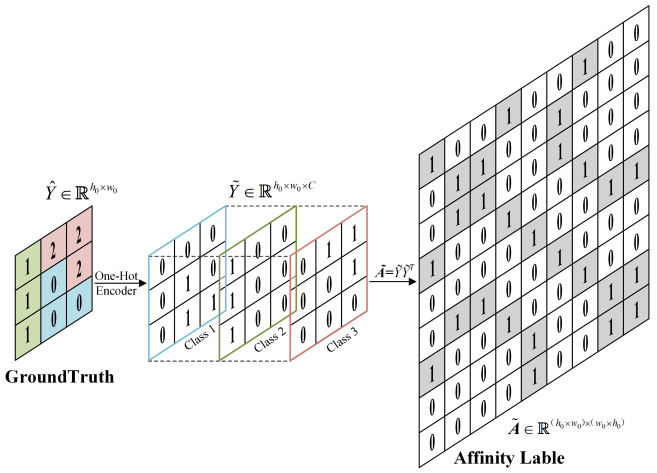


Fig. 4. Illustration of the affinity label generation process, in which the affinity label is a binary result map. A value of 1 indicates that the pixels represented by the corresponding rows and columns belong to the same class (a value of 0 indicates that they belong to different classes).

all the pixels in the feature map $\mathbf{F}_2 \in \mathbb{R}^{h_0 \times w_0 \times c_0}$, and the affinity labels can guide each pixel in the affinity prediction to establish intra- and interclass contextual relationships through supervision by the affinity loss function.

To generate the affinity prediction map, we use a $c_1 \times 1 \times 1$ convolution layer and a BN layer to change the dimensionality of feature map $\mathbf{F}_2$ for obtaining the feature map $\mathbf{F}_3 \in \mathbb{R}^{h_0 \times w_0 \times c_1}$, which is reshaped into a feature map of size $c_1 \times c_1$, and then use a sigmoid function to generate the affinity prediction map $\hat{\boldsymbol{A}} \in \mathbb{R}^{c_1 \times c_1}$, where $c_1 = h_0 \times w_0$.

From the ground truth corresponding to the original HSI, we can know whether a pixel belongs to the same category as other pixels. So, we can generate affinity label maps with intra- and interclass contextual priors based on the ground truth. As shown in Fig. 4, we downsample the size of the ground truth to the same size as $\mathbf{F}_1$ to get a modified ground truth $\hat{Y} \in \mathbb{R}^{h_0 \times w_0}$, and then use a one-hot encoder to encode each label in $\hat{Y}$ and obtain the transformation matrix $\widetilde{Y} \in \mathbb{R}^{h_0 \times w_0 \times C}$, where $C$ denotes the number of classes. After that, we reshape the encoded matrix into $(h_0 \times w_0) \times C$ and multiply it by its transposed matrix to obtain the affinity label map $\widetilde{\boldsymbol{A}}$. The process can be expressed as follows:

$$\widetilde{\boldsymbol{A}} = \widetilde{Y}\widetilde{Y}^{\top} \in \mathbb{R}^{c_1 \times c_1}. \tag{7}$$

The intra- and interclass contextual priors in the affinity labels $\widetilde{\boldsymbol{A}}$ are expressed as binary maps. A value of 1 indicates that the corresponding row and column pixels belong to the same class, while a value of 0 indicates that they belong to different classes. To supervise the affinity prediction map $\hat{\boldsymbol{A}}$ and learn intra- and interclass contextual information from the affinity label map $\widetilde{\boldsymbol{A}}$, it is natural to consider introducing a binary cross entropy loss function to solve the problem as follows:

$$\mathcal{L}_b = -\frac{1}{(c_1)^2} \sum_{i=1}^{(c_1)^2} \left(\widetilde{a}_i \log \hat{a}_i + \left(1 - \widetilde{a}_i\right) \log\left(1 - \hat{a}_i\right)\right) \tag{8}$$

where $\{\hat{a}_i \in \hat{\boldsymbol{A}}, \widetilde{a}_i \in \widetilde{\boldsymbol{A}}, i \in [1, (c_1)^2]\}$. The unary loss ignores the semantic correlation between pixels. Each row of pixels in the affinity prediction map corresponds to a pixel in the feature map $\mathbf{F}_2$. To further improve the semantic correlation between pixels, the pixels in the affinity prediction map $\hat{\boldsymbol{A}}$ are divided into intra- and interclass pixels. The global term of the binary cross entropy loss $\mathcal{L}_g$ is introduced to model the relationship between intra- and interclass contexts, defined as follows:

$$\mathcal{T}_j^p = \log \frac{\sum_{i=1}^{c_1} \widetilde{a}_{ij} \hat{a}_{ij}}{\sum_{i=1}^{c_1} \hat{a}_{ij}} \tag{9}$$

$$\mathcal{T}_j^r = \log \frac{\sum_{i=1}^{c_1} \widetilde{a}_{ij} \hat{a}_{ij}}{\sum_{i=1}^{c_1} \widetilde{a}_{ij}}, \tag{10}$$

$$\mathcal{T}_j^s = \log \frac{\sum_{i=1}^{c_1} \left(1 - \widetilde{a}_{ij}\right)\left(1 - \hat{a}_{ij}\right)}{\sum_{i=1}^{c_1} \left(1 - \widetilde{a}_{ij}\right)} \tag{11}$$

$$\mathcal{L}_g = -\frac{1}{c_1} \sum_{j=1}^{c_1} \left(\mathcal{T}_j^p + \mathcal{T}_j^r + \mathcal{T}_j^s\right) \tag{12}$$

where $\mathcal{T}_j^p$, $\mathcal{T}_j^r$, and $\mathcal{T}_j^s$ denote precision, recall, and specificity, respectively. In affinity loss, the weights $\lambda_b$ and $\lambda_g$ are introduced to balance the unary loss $\mathcal{L}_b$ and the global term $\mathcal{L}_g$, which can be expressed as follows:

$$\mathcal{L}_{\text{aff}} = \lambda_b \mathcal{L}_b + \lambda_g \mathcal{L}_g \tag{13}$$

where $\mathcal{L}_{\text{aff}}$ denotes the affinity loss. In this study, $\lambda_b$ and $\lambda_g$ were set to 1.

### C. RCCA Network

In AL, affinity loss can be used to supervise the affinity prediction map to learn intra- and interclass contextual prior information in the affinity labeling map, which can further refine the global contextual features. As shown in the overall framework presented in Fig. 5, the explicit intraclass context learning can be achieved by applying the affinity prediction map $\hat{\boldsymbol{A}}$ to the feature map $\mathbf{F}_2$, and the process can be expressed as follows:

$$\mathcal{Z}_{\text{tra}} = \hat{\boldsymbol{A}}\mathbf{F}_2' \in \mathbb{R}^{c_1 \times c_0} \tag{14}$$

where $\mathcal{Z}_{\text{tra}}$ denotes the intraclass contextual features and $\mathbf{F}_2'$ denotes the feature map obtained by reshaping $\mathbf{F}_2$ into $c_1 \times c_0 (c_1 = h_0 \times w_0)$. Similarly, by constructing the inverse contextual affinity prediction map, the interclass contextual explicit learning property can be represented as follows:

$$\mathcal{Z}_{\text{ter}} = (\mathbf{1} - \hat{\boldsymbol{A}})\mathbf{F}_2' \in \mathbb{R}^{c_1 \times c_0} \tag{15}$$

where $\mathcal{Z}_{\text{ter}}$ denotes the interclass contextual features and $\mathbf{1}$ represents a matrix of value 1 with the same size as $\hat{\boldsymbol{A}}$. With this intra- and interclass explicit learning, the respective intra- and interclass contexts of pixels in the feature map can be selectively emphasized.

In the classification phase, we obtain the feature maps generated in phase 1 and the contextual features obtained after AL in phase 2, and then fuse them along the channels' dimension. The process can be expressed as follows:

$$\hat{F} = \text{Concat}(\mathbf{F}_2, \mathcal{Z}_{\text{tra}}', \mathcal{Z}_{\text{ter}}') \in \mathbb{R}^{h_0 \times w_0 \times 3c_0} \tag{16}$$

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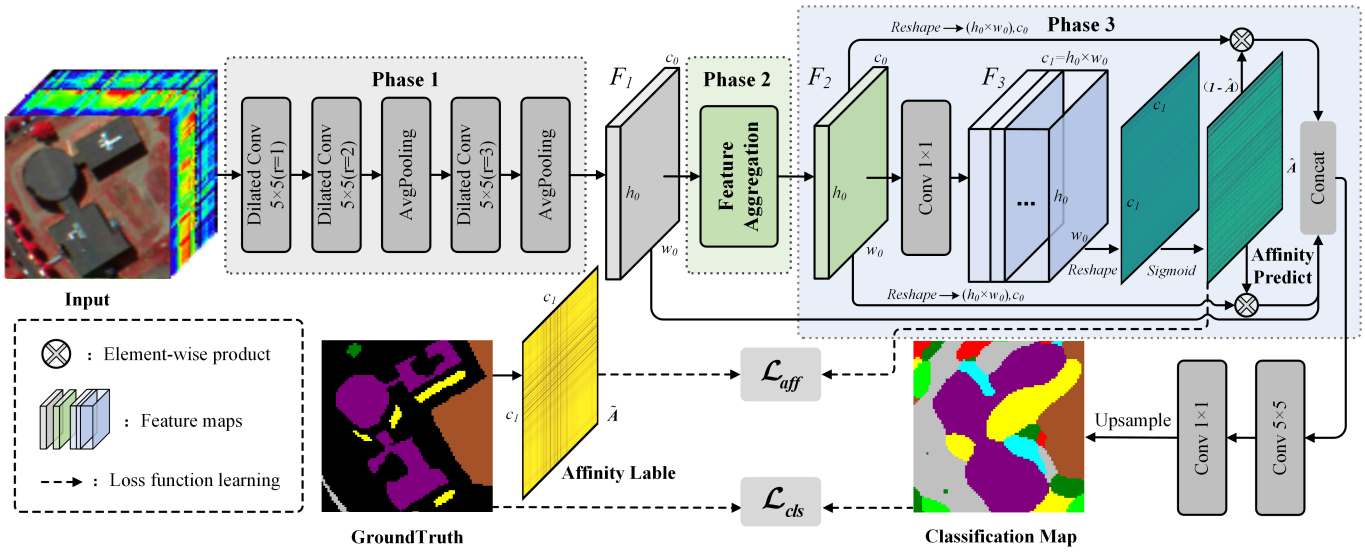


Fig. 5. Architecture of the proposed RCCA. The framework consists of three phases. The first phase involves three convolutional layers with different dilation rates and two pooling layers. In the second phase, an FA module is proposed to model the context of the features obtained by phase 1, which consists of two parallel global and local contextual calibrators. In the third phase, we introduce an AL module (including affinity loss) to distinguish intra- and interclass contextual relationships of the captured contextual features, thereby obtaining more reliable contextual relationships for the final classification.

where $\mathcal{Z}'_{\text{tra}}$ and $\mathcal{Z}'_{\text{ter}}$, respectively denote the feature map obtained by reshaping $\mathcal{Z}_{\text{tra}}$ and $\mathcal{Z}_{\text{ter}}$ into $h_0 \times w_0 \times c_0$, $\hat{F}$ represents the output feature map after the fusion of phase 3, and the final result map is obtained by the convolution layer with softmax and upsampling. Let the prediction result be $\hat{\boldsymbol{Y}}$ and the label be $\boldsymbol{Y}$, the total loss function $\mathcal{L}$ is defined as follows:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{h_0 w_0} \sum_{i=1}^{h_0} \sum_{j=1}^{w_0} \sum_{k=1}^{C} \boldsymbol{Y}_{(i,j,k)} \log\left(\hat{\boldsymbol{Y}}_{(i,j,k)}\right) \tag{17}$$

$$\mathcal{L} = \lambda_{\text{aff}} \mathcal{L}_{\text{aff}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} \tag{18}$$

where $\mathcal{L}_{\text{cls}}$ denotes the segmentation loss, and $\lambda_{\text{aff}}$ and $\lambda_{\text{cls}}$ represent the weights that balance $\mathcal{L}_{\text{aff}}$ and $\mathcal{L}_{\text{cls}}$, respectively. In this study, they were set to 1.

## IV. Experimental Results

This section is structured as follows. First, the datasets used in the experiments are described, followed by the implementation details and a detailed analysis of the experimental results, which consists of three parts. The first part includes ablation studies for analyzing the contribution of the different modules in the proposed method, and the second part involves comparative experiments between the proposed method and several state-of-the-art (SOTA) methods using clean and adversarial test samples. The third part explores the influence of different attack perturbations $\epsilon$ and different attack methods on the classification performance, and finally the complexity of the model is analyzed.

### A. Datasets

Three publicly available datasets were used to verify the validity of the proposed method. These were the Indian Pines [55], Pavia University,[1], and Salinas Valley[1] datasets, which cover rural, urban, and farmland scenes. The corresponding false-color maps and ground-truth maps are shown in Fig. 6. The number of training samples and test samples used in the experiments is reported in Table I.

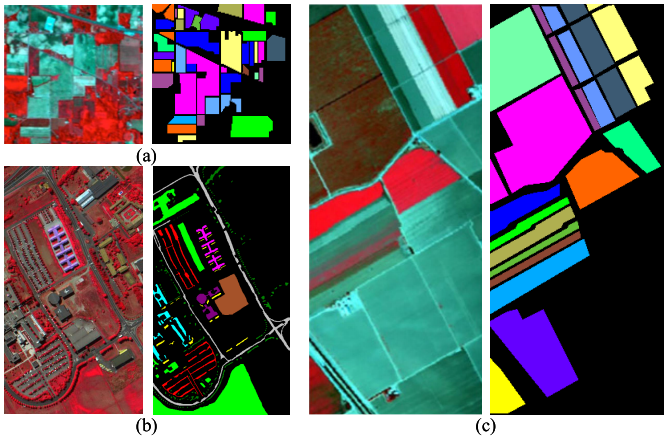


Fig. 6. False-color composite map with the corresponding ground truth for: (a) Indian Pines, (b) Pavia University, and (c) Salinas Valley.

1) *Indian Pines:* This image was taken by the Airborne Visible Infra-Red Imaging Spectrometer (AVIRIS) sensor of NASA's Jet Propulsion Laboratory in the pine forest region of northwestern Indiana, USA. Its size is 145 × 145 pixels, and it includes 200 spectral bands in the wavelength region between 0.4 and 2.5 $\mu$m. There are 10 249 labeled pixels in this image and they belong to 16 land cover classes of forests and crops.
2) *Pavia University:* This image was taken by the reflective optics spectrographic imaging system (ROSIS) of the

[1]http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral Remote Sensing Scenes

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TABLE I
NUMBER OF TRAINING AND TEST SETS CORRESPONDING TO DIFFERENT CLASSES IN THE INDIAN PINES, PAVIA UNIVERSITY, AND SALINAS VALLEY IMAGES

| Indian Pines | | | | | Pavia University | | | | | Salinas Valley | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| No | Name | Training | Test | Toal | No | Name | Training | Test | Toal | No | Name | Training | Test | Toal |
| 1 | Alfalfa | 20 | 26 | 46 | 1 | Asphalt | 100 | 6531 | 6631 | 1 | Weeds_1 | 100 | 1909 | 2009 |
| 2 | Corn-N | 80 | 1348 | 1428 | 2 | Meadows | 100 | 18549 | 18649 | 2 | Weeds_ 2 | 100 | 3626 | 3726 |
| 3 | Corn-M | 80 | 750 | 830 | 3 | Gravel | 100 | 1999 | 2099 | 3 | Fallow | 100 | 1876 | 1976 |
| 4 | Corn | 80 | 157 | 237 | 4 | Tress | 100 | 2964 | 3064 | 4 | Fallow_P | 100 | 1294 | 1394 |
| 5 | Grass-M | 80 | 403 | 483 | 5 | Metal sheets | 100 | 1245 | 1345 | 5 | Fallow_S | 100 | 2578 | 2678 |
| 6 | Grass-T | 80 | 650 | 730 | 6 | Bare soil | 100 | 4929 | 5029 | 6 | Stubble | 100 | 3859 | 3959 |
| 7 | Grass-P | 15 | 13 | 28 | 7 | Bitumen | 100 | 1230 | 1330 | 7 | Celery | 100 | 3479 | 3579 |
| 8 | Hay-W | 80 | 398 | 478 | 8 | Bricks | 100 | 3582 | 3682 | 8 | Grapes | 100 | 11171 | 11271 |
| 9 | Oats | 10 | 10 | 20 | 9 | Shadows | 100 | 847 | 947 | 9 | Soil | 100 | 6103 | 6203 |
| 10 | Soybeans-N | 80 | 892 | 972 | | | | | | 10 | Corn | 100 | 3178 | 3278 |
| 11 | Soybeans-M | 80 | 2375 | 2455 | | | | | | 11 | Lettuce_4wk | 100 | 968 | 1068 |
| 12 | Soybeans-C | 80 | 513 | 593 | | | | | | 12 | Lettuce_5wk | 100 | 1827 | 1927 |
| 13 | Wheat | 80 | 125 | 205 | | | | | | 13 | Lettuce_6wk | 100 | 816 | 916 |
| 14 | Woods | 80 | 1185 | 1265 | | | | | | 14 | Lettuce_7wk | 100 | 970 | 1070 |
| 15 | Buildings | 50 | 336 | 386 | | | | | | 15 | Vinyard_U | 100 | 7168 | 7268 |
| 16 | Stone | 50 | 43 | 93 | | | | | | 16 | Vinyard_T | 100 | 1707 | 1807 |
| | Total | 1025 | 9224 | 10249 | | Total | 900 | 41876 | 42776 | | Total | 1600 | 52529 | 54129 |

German Aerospace Center over the University of Pavia, Italy. Its size in pixels is 610 × 340 and it has 103 spectral bands in the wavelength region between 0.4 and 0.9 $\mu$m. A total of 42 776 pixels are labeled in this image and they belong to nine urban land cover classes.

3) *Salinas Valley:* This image was taken by AVIRIS over the Salinas Valley, California, USA. Its size in pixels is 512 × 217 and includes 204 bands in the wavelength region between 0.2 and 2.5 $\mu$m. There are 54 129 labeled pixels and 16 classes in this image, most of which belong to crops.

### B. Experimental Settings

*1) Settings:* In the following experiments, we use FGSM to simulate the generation of adversarial samples and verify the robustness of the proposed method under adversarial attacks, with the attack perturbation parameter $\epsilon$ set to 0.06. All the experiments were conducted on a computer with an Intel(R) Core(TM) i9-10900KF CPU 3.70 GHz, Nvidia GTX 2080 Ti GPU, and 32 GB of RAM.

*2) Optimized Parameters:* The proposed RCCA was implemented using the PyTorch framework. The training process was optimized using the Adam optimizer, with the learning rate set to 5*e*-4, decay set to 5*e*-5, and number of training epochs set to 500.

*3) Evaluation Metrics:* In our experiments, the overall accuracy (OA), average accuracy (AA), and kappa coefficient were used to evaluate the performance of different classification methods. To reduce the experimental error, all the experiments were repeated ten times, and the mean and variance were reported as the final results.

### C. Ablation Study

In this section, we evaluate the contribution of different modules of the proposed RCCA to the final classification accuracy, including the FA and AL modules. The experimental results on the three datasets mentioned above are shown in Table II, where backbone denotes phase 1 of the proposed framework (which is an FCN composed of dilated convolutions). As it can be seen from the table, applying FA and AL alone achieved higher accuracy in terms of both OA and Kappa compared with the backbone, indicating that the aggregation of spectral and spatial information and the modeling of relationships between intra- and interclass contexts could effectively improve the performance of the model under adversarial attacks. It is worth noting that FA can lead to greater accuracy compared with AL when the two modules are applied alone. Taking the Indian Pines for example, compared with the backbone, FA improved over AL by 2.08% and 2.37% in OA and Kappa, respectively. This trend is also evident in the Pavia University and Salinas Valley. This shows that contextual FA has a significant effect on modeling intra- and interclass contextual relationships. The experimental results obtained by applying both FA and AL further validated our judgment. Compared with the backbone, applying AL after FA can greatly improve the classification performance of the model under adversarial samples. In the case of Pavia University, the improvements in accuracy obtained by applying FA and AL alone were no more than 5%, while applying them together produced 10.74% and 13.97% improvements in terms of OA and Kappa. This suggests that capturing intra- and interclass contextual relationships by learning prior knowledge of class contextual information after FA can effectively suppress the effect of unreliable contextual information, and thus improve the classification performance of the model in the presence of adversarial samples.

### D. Comparison With Other Methods

In this section, we compare the proposed method with several SOTA methods, including 3-DCNN [39],

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TABLE II

PERFORMANCE CONTRIBUTION OF DIFFERENT MODULES IN THE PROPOSED RCCA ON THREE DATASETS, WHICH INCLUDE FA MODULE AND AL MODULE. THE VALUES IN BRACKETS REPRESENT THE GAP IN ACCURACY WITH THE BACKBONE, AND THE NUMBERS IN BOLD INDICATE THE BEST VALUE. ALL THE RESULTS ARE IN %

| Method | FA | AL | Indian Pines | | Pavia University | | Salinas Valley | |
|---|---|---|---|---|---|---|---|---|
| | | | OA | Kappa | OA | Kappa | OA | Kappa |
| Backbone | - | - | 87.28 | 85.53 | 78.70 | 72.06 | 75.72 | 73.04 |
| +FA | ✓ | - | 94.44(↑7.16) | 93.63(↑8.10) | 81.74(↑3.04) | 76.31(↑4.25) | 79.66(↑3.95) | 77.36(↑4.32) |
| +AL | - | ✓ | 92.36(↑5.08) | 91.26(↑5.73) | 80.71(↑2.10) | 75.55(↑3.49) | 78.65(↑2.93) | 76.07(↑3.03) |
| +FA&AL | ✓ | ✓ | **95.63(↑8.35)** | **93.75(↑8.22)** | **89.44(↑10.74)** | **86.03(↑13.97)** | **86.54(↑10.82)** | **84.89(↑11.85)** |

spectral–spatial residual network (SSRN) [56], spectral–spatial FCN (SSFCN) [29], and self-attention contextual network (SACNet) [38]. In these methods, the specific network structures and parameter settings can be found in the corresponding references. The test samples included clean and adversarial samples. The settings of adversarial samples are shown in part B(1) of this section, and the number of training and test sets is shown in Table I.

*1) Experimental Results With Indian Pines:* The quantitative and qualitative results with the Indian Pines image are reported in Table III and Fig. 7. The accuracy of most classification methods that performed well with clean samples was greatly decreased when adversarial samples were introduced. The SSRN (with the best performance in the clean dataset) showed a decrease in OA of 61.75% and a decrease in AA of 73.14% with adversarial samples. The accuracy of some classes, such as Alfalfa, Hay-W, and Moth C, was close to 0, which indicates that most advanced deep-learning-based HSI classification methods are very vulnerable to adversarial attacks. SACNet and the proposed RCCA obtained the best performance with adversarial samples. This suggests that by focusing on the global contextual information of pixels, the negative impact of adversarial samples can be controlled. Compared with SACNet, the proposed method improved OA by 5.26% and 6.7% in the non-attacked and attacked states, respectively, indicating that refining the global context information by distinguishing intraclass context information and interclass context information can improve the robustness of the model against adversarial samples. From the classification maps presented in Fig. 7, it can be seen that the performance of the proposed method on both clean and attacked samples was consistent with the ground truth in the Hay-W and Oats classes, which further reflected the advantages of the proposed method in the presence of adversarial samples.

*2) Experimental Results With Pavia University:* Unlike the Indian Pines dataset, the ground objects in this image are mostly urban, and their distribution is relatively dense, which increases the difficulty of classification. The experimental results of different methods with the Pavia University are shown in Table IV. The accuracy obtained by 3-DCNN, SSRN, and SSFCN with clean and adversarial samples was very different, which indicates that classification methods are vulnerable to adversarial samples. SACNet and the proposed RCCA obtained satisfactory accuracy with adversarial

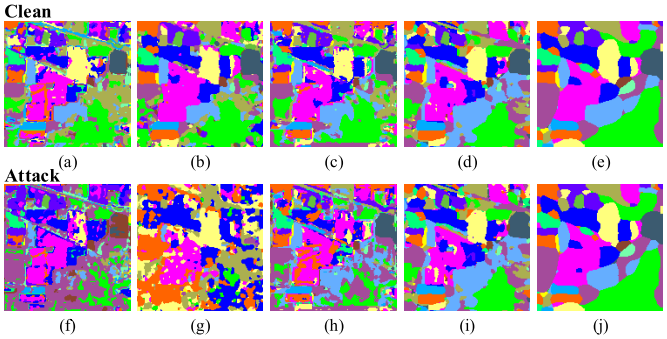


Fig. 7. Classification maps obtained by testing with clean samples (first column) and with adversarial samples (second column) in the Indian Pines image. **Clean samples**: (a) 3-DCNN (79.01), (b) SSRN (96.50), (c) SSFCN (85.76), (d) SACNet (91.00), (e) RCCA (96.26); **Adversarial samples**: (f) 3-DCNN (56.45), (g) SSRN (34.75), (h) SSFCN (61.11), (i) SACNet (88.93), and (j) RCCA (95.63). The numbers in brackets represent the OAs corresponding to different methods (in %).

samples, indicating that focusing on the global contextual information by capturing the long-range dependencies of pixels is conducive to alleviating the negative impact of adversarial samples. As shown in Fig. 8, regardless of the presence of clean or adversarial samples, many fewer classes were misclassified by the proposed RCCA than by other methods, thus obtaining performance that is closer to those with the real distribution. This finding suggests that by focusing on the intra- and interclass contexts, the proposed method can refine the global contextual information and improve performance in the presence of adversarial attacks.

*3) Experimental Results With Salinas Valley:* This dataset is similar to the Indian Pines dataset, while the ground object distribution in Salinas Valley is more smooth and continuous. The experimental results obtained by the different methods are shown in Table V. Consistent with the observations in the previous two datasets, most classification methods were severely affected by the presence of adversarial samples. We also found that although advanced classification methods such as SSRN performed best in clean samples, they were vulnerable to adversarial samples and only obtain 32.99% in OA. This indicates that advanced methods are more sensitive to adversarial samples. Another point of concern is that classification methods such as 3-DCNN and SSRN exhibit high standard deviations when tested with adversarial samples, indicating that adversarial attacks can affect the stability of classification

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TABLE III

CLASSIFICATION ACCURACY OBTAINED BY DIFFERENT METHODS TESTED WITH CLEAN AND ADVERSARIAL SAMPLES IN THE INDIAN PINES IMAGE. THE NUMBER OF TRAINING AND TESTING SAMPLES IS SHOWN IN TABLE I. THE NUMBERS IN BRACKETS INDICATE THE CORRESPONDING STANDARD DEVIATIONS, AND THE NUMBERS IN BOLD INDICATE THE BEST VALUES

| Class | Test with clean samples | | | | | Test with adversarial samples | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 3DCNN | SSRN | SSFCN | SACNet | RCCA | 3DCNN | SSRN | SSFCN | SACNet | RCCA |
| Alfalfa | 88.08 | 99.62 | 90.00 | 96.15 | 94.23 | 98.46 | 3.08 | 92.31 | 95.38 | 87.69 |
| Corn-N | 68.06 | 94.31 | 82.79 | 89.29 | 96.60 | 60.05 | 68.98 | 60.82 | 89.22 | 97.43 |
| Corn-M | 69.20 | 97.00 | 85.44 | 82.12 | 96.33 | 27.56 | 83.77 | 73.44 | 79.97 | 95.33 |
| Corn | 91.08 | 99.62 | 97.20 | 98.85 | 99.68 | 71.40 | 2.80 | 94.27 | 98.34 | 98.09 |
| Grass-M | 94.42 | 98.61 | 95.78 | 97.07 | 98.91 | 48.88 | 10.27 | 89.08 | 96.30 | 98.56 |
| Grass-T | 98.25 | 99.63 | 94.65 | 98.14 | 99.52 | 51.63 | 18.95 | 88.18 | 96.54 | 97.72 |
| Grass-P | 100.00 | 100.00 | 98.46 | 100.00 | 100.00 | 94.62 | 3.08 | 98.46 | 98.46 | 98.46 |
| Hay-W | 97.86 | 100.00 | 98.19 | 97.06 | 100.00 | 32.16 | 1.81 | 97.59 | 97.06 | 100.00 |
| Oats | 87.00 | 100.00 | 84.00 | 100.00 | 100.00 | 77.00 | 15.00 | 86.00 | 100.00 | 100.00 |
| Soybeans-N | 78.81 | 95.09 | 85.49 | 90.67 | 96.82 | 30.46 | 68.49 | 32.67 | 86.39 | 96.08 |
| Soybeans-M | 65.55 | 93.88 | 74.15 | 86.52 | 92.40 | 78.76 | 15.42 | 38.27 | 84.35 | 91.01 |
| Soybeans-C | 81.62 | 97.90 | 86.16 | 86.86 | 92.26 | 64.37 | 1.13 | 68.23 | 86.08 | 92.44 |
| Wheat | 99.36 | 99.84 | 99.04 | 98.56 | 99.20 | 71.92 | 1.92 | 98.40 | 87.92 | 99.20 |
| Woods | 94.87 | 99.16 | 95.90 | 97.32 | 99.05 | 54.89 | 20.68 | 67.70 | 93.55 | 98.60 |
| Buildings | 85.03 | 98.60 | 88.15 | 97.47 | 99.55 | 35.06 | 68.36 | 69.88 | 95.33 | 99.58 |
| Stone | 98.60 | 100.00 | 98.60 | 98.84 | 100.00 | 97.44 | 19.30 | 96.28 | 96.74 | 99.07 |
| OA | 79.01 (±0.99) | **96.50 (±0.45)** | 85.76 (±0.91) | 91.00 (±0.84) | 96.26 (±0.43) | 56.45 (±4.94) | 34.75 (±9.27) | 61.11 (±4.81) | 88.93 (±2.25) | **95.63 (±3.53)** |
| AA | 87.36 (±0.81) | **98.33 (±0.27)** | 90.88 (±1.04) | 94.68 (±0.66) | 97.79 (±0.26) | 62.17 (±7.06) | 25.19 (±5.43) | 78.22 (±2.21) | 92.60 (±1.89) | **96.83 (±1.05)** |
| kappa | 76.14 (±1.11) | **95.97 (±0.52)** | 83.75 (±1.03) | 89.66 (±0.96) | 95.70 (±0.37) | 49.63 (±5.71) | 26.96 (±9.52) | 56.39 (±4.87) | 87.29 (±2.57) | **94.97 (±3.97)** |

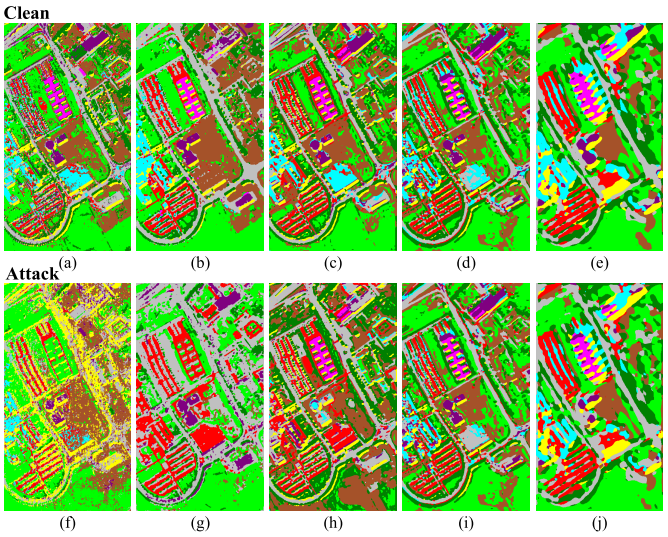


Fig. 8. Classification maps obtained by testing with clean samples (first column) and testing with adversarial sample (second column) in the Pavia University image. **Clean samples**: (a) 3-DCNN (85.30), (b) SSRN (98.34), (c) SSFCN (93.84), (d) SACNet (88.53), (e) RCCA (91.08); **Adversarial samples**: (f) 3-DCNN (45.49), (g) SSRN (60.88), (h) SSFCN (45.65), (i) SACNet (87.22), and (j) RCCA (89.44). The numbers in brackets represent the OAs corresponding to different methods (in %).

methods. The proposed RCCA achieved satisfactory accuracy on both clean and adversarial samples and had low standard deviation on adversarial samples, reflecting the robustness of the proposed method under adversarial attacks. As shown in Fig. 9, the SOTA methods achieved poor classification performance when tested with adversarial samples, while the proposed RCCA obtained the best classification results by distinguishing intra- and interclass contextual information.

### *E. Influence of Different Perturbations and Attack Methods*

In this section, we first evaluate the influence of attack perturbations with different strengths on classification accuracy, by setting the $\epsilon$ range as [0.01–0.1] with a step size of 0.01. The experiment compared five advanced classification methods on the three aforementioned HSI datasets. As shown in Fig. 10, with an increase in attack disturbance, the classification accuracy of the different methods showed a decreasing trend. This indicates that the larger the perturbation, the less robust the classification model with adversarial samples. The classification performance of the model can be controlled by modifying the intensity of the perturbation, which further reflects the vulnerability of deep learning models for HSI classification. We also found that the classification model performed worse under adversarial samples in datasets with smoother ground object distributions, such as Indian Pines and Salinas Valley. Taking the excellent classification method SSRN for example, when the attack disturbance $\epsilon$ was set 0.01, the OA value dropped below 50% in Indian Pines and Salinas Valley, while the accuracy could reach about 75% in Pavia University. This indicates that scenarios with more complex distribution of ground objects were conducive to alleviating the threat caused by adversarial attacks. The accuracy of SACNet and the proposed RCCA did not change significantly as the attack perturbation strength increases. This indicates that the threat

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TABLE IV

CLASSIFICATION ACCURACY OBTAINED BY DIFFERENT METHODS TESTED WITH CLEAN AND ADVERSARIAL SAMPLES IN THE PAVIA UNIVERSITY IMAGE. THE NUMBER OF TRAINING AND TESTING SAMPLES IS SHOWN IN TABLE I. THE NUMBERS IN BRACKETS INDICATE THE CORRESPONDING STANDARD DEVIATIONS, AND THE NUMBERS IN BOLD INDICATE THE BEST VALUES

| Class | Test with clean samples | | | | | Test with adversarial samples | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 3DCNN | SSRN | SSFCN | SACNet | RCCA | 3DCNN | SSRN | SSFCN | SACNet | RCCA |
| Asphalt | 86.28 | 98.41 | 89.51 | 83.85 | 84.03 | 3.40 | 47.60 | 95.51 | 84.03 | 85.12 |
| Meadows | 88.51 | 98.76 | 95.19 | 87.98 | 92.12 | 71.56 | 98.70 | 13.66 | 86.81 | 92.60 |
| Gravel | 72.85 | 93.65 | 92.97 | 87.40 | 97.45 | 17.31 | 19.03 | 4.13 | 85.20 | 85.44 |
| Tress | 89.02 | 98.69 | 97.56 | 92.30 | 74.97 | 80.18 | 36.92 | 98.47 | 92.05 | 73.68 |
| Metal sheets | 99.28 | 100.00 | 99.86 | 98.60 | 98.55 | 40.05 | 6.66 | 99.71 | 98.78 | 95.34 |
| Bare soil | 76.81 | 98.88 | 91.39 | 87.36 | 97.57 | 15.27 | 16.96 | 49.09 | 84.90 | 95.01 |
| Bitumen | 84.31 | 99.11 | 95.56 | 94.13 | 96.99 | 42.52 | 40.34 | 15.98 | 79.43 | 64.47 |
| Bricks | 74.81 | 96.40 | 91.43 | 91.32 | 94.39 | 29.39 | 33.12 | 73.67 | 91.73 | 94.64 |
| Shadows | 98.54 | 99.95 | 99.83 | 98.11 | 92.44 | 0.69 | 0.02 | 99.76 | 97.52 | 91.26 |
| OA | 85.30 (±3.52) | **98.34 (±0.36)** | 93.84 (±0.29) | 88.53 (±3.17) | 91.08 (±1.12) | 45.49 (±5.22) | 60.88 (±9.61) | 45.65 (2.56) | 87.22 (±3.10) | **89.44 (±1.63)** |
| AA | 85.60 (±5.35) | **98.20 (±0.27)** | 94.81 (±0.32) | 91.23 (±1.43) | 92.06 (±0.70) | 33.37 (±7.53) | 33.26 (±12.19) | 61.11 (±3.82) | **88.94 (±3.98)** | 86.40 (±3.23) |
| Kappa | 80.61 (±4.73) | **97.79 (±0.47)** | 91.83 (±0.39) | 85.01 (±3.94) | 88.25 (±1.43) | 31.62 (±6.32) | 38.35 (±19.9) | 37.63 (±2.15) | 83.30 (±3.91) | **86.03 (±2.12)** |

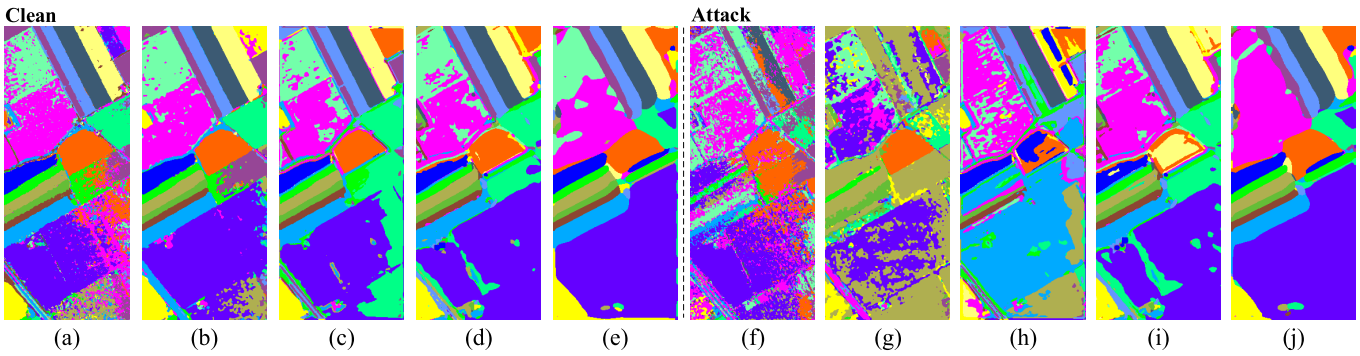


Fig. 9. Classification maps obtained by testing with clean samples (left of dividing line) and by testing with adversarial samples (right of dividing line) in the Salinas Valley image. **Clean samples**: (a) 3-DCNN (94.29), (b) SSRN (97.65), (c) SSFCN (92.38), (d) SACNet (92.30), (e) RCCA (95.02); **Adversarial samples**: (f) 3-DCNN (54.25), (g) SSRN (32.99), (h) SSFCN (44.35), (i) SACNet (84.45), and (j) RCCA (86.54). The numbers in brackets represent the OA corresponding to different methods (in %).

of adversarial attack in HSI classification can be effectively resisted by focusing on the global context. When the attack perturbation parameter $\epsilon$ reached 0.1, RCCA still obtained the best performance on the three datasets, further demonstrating the robustness of the proposed method to adversarial samples.

Another experiment is to analyze the performance of the proposed model under different attack models. In addition to FGSM, we also compare the classification performance of the different methods under PGD and C&W attack. The details of these two attack methods are described in Section II-A. For the setting in PGD and C&W settings, the perturbation is kept the same as the FGSM at 0.06 and the number of iterations is set to 5 to generate the adversarial samples. From the results presented in Table VI, the accuracy obtained by different methods under PGD and C&W attacks in the three datasets is lower than FSGM, which reflects that PGD and C&W are more threatening to the deep model than FGSM. The proposed RCCA maintains the best performance in both farmland and urban scenes under three different attack modes, which validates the robustness and generalization of the proposed method for HSI classification.

### F. Analysis of Model Complexity

In this section we, analyze the complexity of the model. Table VII shows the running times of different methods trained once on the three datasets. In general, the computational complexity of the model had a positive correlation with the structure and depth of the network. Compared with the image-based methods (such as SSFCN, SACNet, and RCCA), the patch-based methods (3-DCNN and SSRN) needed more time to be trained and tested. On one hand, the network is deep, and on the other hand, the patch-based methods lead to patch overlaps between adjacent pixels, which reduces the speed of model inference. Although the running time of our proposed RCCA was higher than other image-based methods, its testing time was almost negligible, which is important for practical application of the model in scenarios that allow offline training and online real-time inference.

## V. PRACTICAL APPLICATION

We used the GF-5 satellite dataset to further verify the robustness of the proposed method to adversarial samples in

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TABLE V

CLASSIFICATION ACCURACY OBTAINED BY DIFFERENT METHODS TESTED WITH CLEAN AND ADVERSARIAL SAMPLES IN THE SALINAS VALLEY IMAGE. THE NUMBER OF TRAINING AND TESTING SAMPLES IS SHOWN IN TABLE I. THE NUMBERS IN BRACKETS INDICATE THE CORRESPONDING STANDARD DEVIATIONS, AND THE NUMBERS IN BOLD INDICATE THE BEST VALUES

| Class | Test with clean samples | | | | | Test with adversarial samples | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 3DCNN | SSRN | SSFCN | SACNet | RCCA | 3DCNN | SSRN | SSFCN | SACNet | RCCA |
| Weeds_1 | 99.70 | 100.00 | 99.95 | 99.99 | 99.95 | 33.85 | 1.19 | 100.00 | 97.15 | 99.89 |
| Weeds_ 2 | 99.32 | 99.99 | 98.57 | 97.95 | 99.32 | 90.34 | 68.96 | 46.11 | 64.46 | 99.67 |
| Fallow | 99.13 | 100.00 | 99.72 | 98.35 | 99.90 | 52.40 | 36.53 | 0.03 | 89.55 | 99.42 |
| Fallow_P | 99.20 | 99.91 | 99.88 | 97.18 | 98.04 | 38.90 | 15.76 | 33.32 | 95.60 | 87.69 |
| Fallow_S | 98.99 | 99.62 | 99.78 | 98.16 | 98.12 | 34.18 | 6.68 | 28.23 | 75.68 | 94.21 |
| Stubble | 99.94 | 100.00 | 99.77 | 99.68 | 98.02 | 35.76 | 0.05 | 92.00 | 98.81 | 98.87 |
| Celery | 99.21 | 99.99 | 98.30 | 96.80 | 98.77 | 50.15 | 7.48 | 7.25 | 96.82 | 98.16 |
| Grapes | 87.65 | 93.96 | 78.55 | 82.21 | 85.27 | 49.57 | 6.12 | 70.37 | 80.39 | 93.59 |
| Soil | 99.54 | 99.99 | 98.83 | 96.90 | 99.58 | 81.69 | 58.43 | 2.52 | 87.96 | 97.70 |
| Corn | 96.80 | 99.14 | 92.70 | 97.76 | 99.67 | 53.28 | 22.09 | 78.24 | 97.36 | 98.51 |
| Lettuce_4wk | 99.71 | 99.98 | 97.09 | 98.44 | 98.37 | 68.78 | 27.49 | 63.31 | 98.61 | 98.74 |
| Lettuce_5wk | 99.07 | 100.00 | 100.00 | 100.00 | 99.66 | 63.12 | 68.93 | 13.14 | 94.69 | 98.28 |
| Lettuce_6wk | 99.41 | 99.99 | 99.49 | 99.68 | 99.07 | 34.53 | 12.27 | 78.60 | 99.69 | 97.56 |
| Lettuce_7wk | 98.43 | 99.74 | 99.92 | 98.66 | 99.79 | 61.09 | 1.79 | 78.87 | 99.54 | 97.02 |
| Vinyard_U | 81.63 | 92.84 | 84.11 | 80.02 | 90.64 | 42.76 | 92.29 | 17.30 | 64.38 | 21.71 |
| Vinyard_T | 98.69 | 99.75 | 99.98 | 98.79 | 99.26 | 64.91 | 16.12 | 44.02 | 93.79 | 98.80 |
| OA | 94.29 (±0.51) | **97.65 (±0.40)** | 92.38 (±0.38) | 92.30 (±1.86) | 95.02 (±0.70) | 54.25 (±14.83) | 32.99 (±7.34) | 44.35 (±2.92) | 84.45 (±3.13) | **86.54 (±1.54)** |
| AA | 97.28 (±0.31) | **99.06 (±0.14)** | 96.66 (±0.11) | 96.29 (±0.69) | 97.72 (±0.23) | 53.46 (±15.49) | 27.64 (±8.52) | 47.08 (±5.24) | 89.65 (±2.80) | **92.49 (±1.35)** |
| kappa | 93.62 (±0.57) | **97.38 (±0.45)** | 91.52 (±0.42) | 91.43 (±2.07) | 94.45 (±0.78) | 48.94 (±16.24) | 25.68 (±8.99) | 38.27 (±3.12) | 82.70 (±3.48) | **84.89 (±1.71)** |

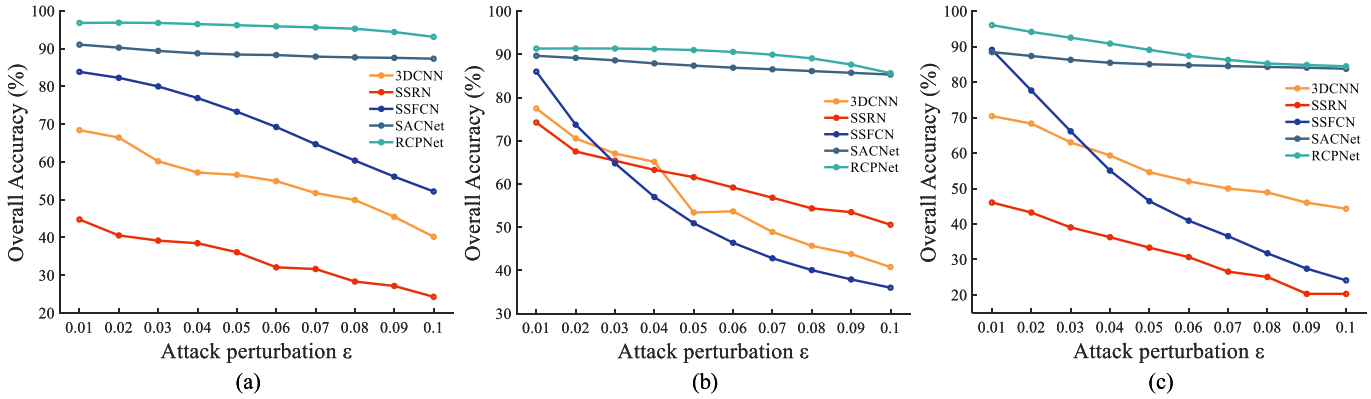


Fig. 10. Influence of different values of the attack perturbation parameter $\epsilon$ on classification performance. (a) Indian Pines, (b) Pavia University, (c) Salinas Valley.

practical applications. The GF-5 satellite was launched on May 9, 2018, and the HSI used in our experiments was taken by the satellite on January 22, 2019. GF-5 carries six payloads and can observe both the atmosphere and the land simultaneously [57]. The obtained HSI images have 330 spectral channels with spectral resolution up to 5 nm and wavelength range of 0.4–2.5 $\mu$m. The experimental area is located in the area around the East Dongting Lake in Yueyang, Hunan Province, China. Fig. 11(d) shows a false-color composite map of the scene, which has a spatial size of 456 $\times$ 352 and 310 spectral bands for analysis. Our study combined the actual situation of the region with the visual interpretation of high-resolution remote sensing images in Google Earth to label the original image. There were 4816 labeled samples in this scene, which were classified into six different classes. Fig. 11(h) shows the ground truth of this image. We performed preprocessing operations such as atmospheric correction and radiometric calibration on the original image before the experiment.

In our experiments, we randomly selected 12 samples for training, and the adversarial samples were generated by FGSM with $\epsilon = 0.1$. Fig. 11 shows the classification results obtained by different methods in the GF-5 dataset. There was a large gap between the obtained accuracies with clean and adversarial samples, such as the difference of 16.13% between the accuracies obtained by SSFCN with clean and adversarial samples. This result indicated that adversarial attacks had a significant influence on the deep-learning-based model in practical HSI

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TABLE VI

CLASSIFICATION PERFORMANCE OF ALL COMPARISON METHODS UNDER DIFFERENT ATTACK MODES IN THREE DATASETS. THE RESULTS ARE PRESENTED IN OA (IN%). BOLD INDICATES THE BEST VALUE

| Indian Pines | | | | | |
|---|---|---|---|---|---|
| Method | 3DCNN | SSRN | SSFCN | SACNet | RCCA |
| FGSM | 56.45 | 34.75 | 61.11 | 88.93 | **95.63** |
| PGD | 44.83 | 58.15 | 52.56 | 78.90 | **87.98** |
| C&W | 20.93 | 23.18 | 18.38 | 79.15 | **88.65** |
| Pavia University | | | | | |
| Method | 3DCNN | SSRN | SSFCN | SACNet | RCCA |
| FGSM | 45.49 | 60.88 | 45.65 | 87.22 | **89.44** |
| PGD | 20.12 | 53.62 | 32.94 | 76.05 | **82.81** |
| C&W | 36.66 | 52.36 | 42.30 | 71.60 | **74.71** |
| Salinas Valley | | | | | |
| Method | 3DCNN | SSRN | SSFCN | SACNet | RCCA |
| FGSM | 54.25 | 32.99 | 44.35 | 84.45 | **86.54** |
| PGD | 46.78 | 30.04 | 40.27 | 74.42 | **86.60** |
| C&W | 43.56 | 19.93 | 16.18 | 72.58 | **77.13** |

TABLE VII

RUNNING TIME OF ALL CLASSIFICATION METHODS ON THE THREE BENCHMARK DATASETS (IN SECONDS)

| Method | Indian Pines | | Pavia University | | Salinas Valley | |
|---|---|---|---|---|---|---|
| | Train | Test | Train | Test | Train | Test |
| 3DCNN | 256.40 | 1.51 | 335.64 | 6.38 | 341.93 | 4.65 |
| SSRN | 349.15 | 3.62 | 396.49 | 9.11 | 418.81 | 6.32 |
| SSFCN | 6.63 | 0.01 | 51.61 | 0.08 | 36.61 | 0.06 |
| SACNet | 8.89 | 0.03 | 70.06 | 0.09 | 38.58 | 0.06 |
| RCCA | 8.24 | 0.02 | 181.59 | 0.17 | 59.14 | 0.09 |

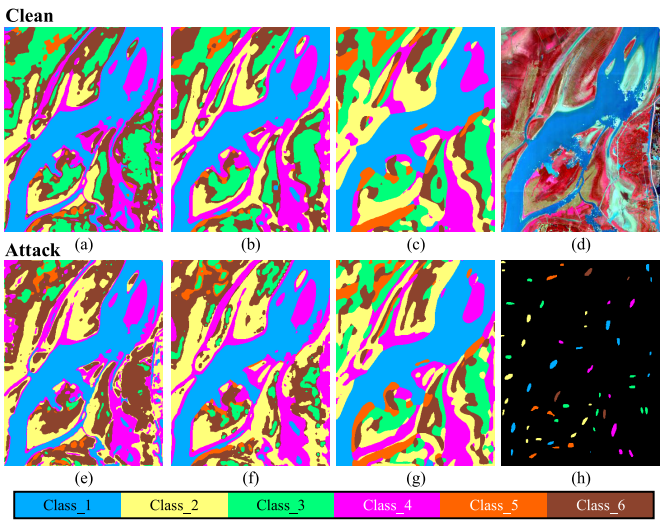


Fig. 11. Classification maps obtained by testing with clean samples (first column) and testing with adversarial sample (second column) in the GF-5 dataset. **Clean samples**: (a) SSFCN (93.42), (b) SACNet (85.78), (c) RCCA (89.35), and (d) false-color composite map; **Adversarial samples**: (e) SSFCN (77.29); (f) SACNet (84.19), (g) RCCA (86.27), and (h) ground truth. The numbers in brackets represent the OAs corresponding to different methods (in %).

classification application. Although the proposed RCCA had a 4.07% difference in OA compared with SSFCN with clean samples, it produced an 8.98% improvement compared with SSFCN when confronted with adversarial attacks, indicating that the proposed RCCA can achieve a good balance between generalization and robustness in practice.

## VI. CONCLUSION

We introduce a new RCCA framework for HSI classification which specifically addresses the vulnerability of the model to adversarial samples in practical applications. Our proposed method models the global context relationships between pixels in the whole image by feature extraction and aggregation, and then distinguishes intra- and interclass contextual information by supervised affinity loss to obtain finer global contextual features, thus improving the robustness of the model to adversarial samples. Experiments on three benchmark datasets and a GF-5 dataset demonstrate that the proposed method achieves a good balance between accuracy and robustness in the presence of adversarial samples.

The proposed HSI defense model focuses on white-box attacks. However, most attacks in real scene are black-box attacks because the attacker does not have access to the detailed parameters of the model. Therefore, improving the robustness of the model under black-box attacks is our future research topic. Besides, transformer has also achieved impressive performance in HSI processing, and its robustness is an interesting research issue. We will investigate this issue in our future work.

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